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LETTER

A Low-Power Keyword Spotting Chip with Multiplier-Free MFCC Feature Extractor

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Abstract Mel-frequency cepstral coefficients (MFCC), an FFT-based speech feature extraction (FEx) algorithm, is a significant power consumer in low-power keyword spotting (KWS) chips. This work presents a KWS chip with an energy-efficient FEx, with an expanded-3bit-twiddle FFT (E3bT-FFT) algorithm which reduces power of FFT by 5.7x. Meanwhile, a multiplier-free MFCC (MF-MFCC) is proposed, effectively eliminating power-hungry multipliers and reducing the MFCC computational load by 7.3x. Fabricated in a 65-nm CMOS process, the chip occupies 0.17 mm^2 and consumes 2.3 μW, with the computation unit in FEx consuming just 76 nW, and achieves 94.9% accuracy on a 1-Word KWS with Google Speech Commands dataset (GSCD).

Keywords: Fast Fourier transform (FFT), keyword spotting (KWS), Mel Frequency Cepstral Coefficients (MFCC), neural network (NN).
Classification: Devices, circuits and hardware for IoT and biomedical applications

1. Introduction

Recently neural network algorithms are attractive in natural language process, promoting an increased demand of low-power voice command methods for mobile and Internet of Things (IoT) devices [1-4]. In such devices, the keyword spotting (KWS) expolited to receive human commands generally dominate system power consumption [5-12], and then requiring low-power solution.

The typical KWS algorithm, as shown in Fig. 1, mainly includes Mel Frequency Cepstral Coefficients (MFCC) and neural network classifier. To address the challenge of reducing the power consumption of KWS, three primary methods have been developed in recent years. The first method seeks to minimize the running time of KWS by implementing multi-stage KWS with a voice activity detector (VAD) and incorporating regular sleep and wake-up cycles [13,14]. The second method focuses on decreasing the power consumption of the classiffer by training the network with quantization and pruning techniques, or using a more efffcient neural network architecture [15,16,35]. The third method aims to reduce the power consumption of FEx by replacing MFCC with an analog signal processor (ASP), such as analog bandpass ffilters [17-34]. But, this method increase the chip area and decrease algorithm accuracy. The works in [14, 15] utilize digital MFCC as a feature extractor. Compared to analog feature extractors (FEx), MFCC is easier to deploy on IoT devices as it does not require custom analog circuitry. However, the power consumption of MFCC remains a significant challenge. Although these works leverage advanced techniques to optimize and reduce the MFCC operation time, MFCC is still the primary factor affecting power consumption.

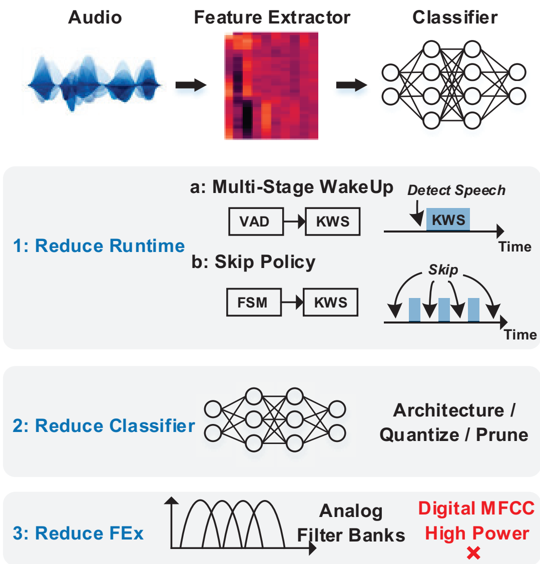

Fig. 1 The overview of KWS, and three methods to decrease the power consumption.

To address these issues, this work presents a low-power KWS chip featuring a multiplier-free MFCC (MF-MFCC). A low-power FFT algorithm with expanded 3-bit twiddle (E3bT) is introduced, replacing the multiplier in butterfly operations with addition and shift operations. The MF-MFCC is further optimized to enhance processing efficiency, with a detailed performance analysis provided. Additionally, a configurable multiplier-free processing element (PE) is implemented in the MFCC processor, allowing power, Mel-filter, and DCT operations to share the same PE, reducing both power and area consumption.

The remains of this brief are organized as follows. Section II introduces the algorithm of MF-MFCC and the neural network classifier for KWS. Section III illustrates the overall hardware architecture, including the MF-MFCC processor with E3bT butterfly unit and configurable multiplier-free PE, and the deep learning accelerator with binary PE. Section IV provides evaluations, measurement results, and comparisons. Finally, conclusions are drawn in Section V.

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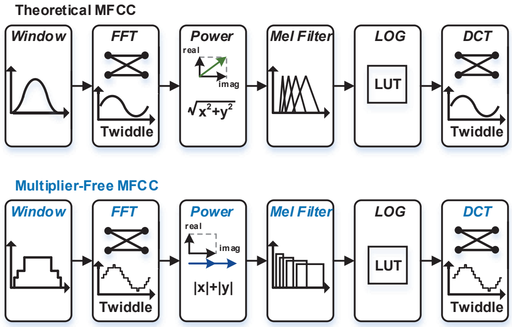


Fig. 2 The architecture of the conventional MFCC and MF-MFCC.

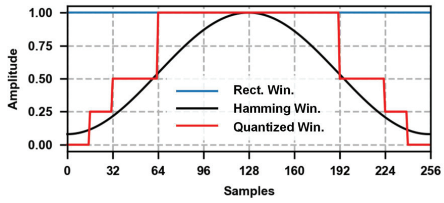


Fig. 3 The window function of Hamming window, rectangle window and quantized window.

2. Algorithm

The MFCC algorithm, as shown in Fig. 2, maps the FFT spectrum to a range more aligned with human perception using a Mel filter bank, thereby improving the performance of speech algorithms.

Before performing FFT operations, the audio in each frame is weighted with the Hamming window function to prevent frequency leakage. To reduce the computational complexity of the Hamming window, this paper introduces a quantization method, as shown in (1). By converting the original Hamming window from floating-point to discrete values, the multiplication is replaced with shift operations. Additionally, the quantized window function eliminates the need to store window function parameters, reducing memory power consumption and area.

$$W(x) = \begin{cases} 0, & \left|idx - \frac{N}{2}\right| > \frac{15N}{16} \\ 0.25, & \frac{15N}{16} > \left|idx - \frac{N}{2}\right| >= \frac{3N}{8} \\ 0.5, & \frac{3N}{8} > \left|idx - \frac{N}{2}\right| >= \frac{N}{4} \\ 1, & \left|idx - \frac{N}{2}\right| < \frac{N}{4} \end{cases} \tag{1}$$

The FFT is a key component of the MFCC, converting audio from the time domain to the frequency domain. The Radix-2 real FFT transforms real-number audio signals into complex numbers and performs butterfly operations. The butterfly operation involves complex multiplication and addition between the input and Twiddle, consisting of four real-number multiplications and six additions. As shown in Fig. 4(a), we quantize the Twiddle factors and extend one sign bit for Twiddles smaller than 4 bits to represent the positive 1.0. We conducted a performance analysis of the FFT using quantized Twiddle in butterfly operations. By applying FFT to a combination of sine waves across different frequency bands, we tested the FFT's resolution for signals within the 0-2000 Hz range, which covers the primary frequency range of human hearing. As shown in Fig. 4(c), the FFT with E3bT maintains accurate performance within the 0-1000 Hz range but shows reduced signal resolution in the 1000-2000 Hz range. In contrast, as shown in Fig. 4(d), the FFT with E2bT experiences a rapid decline in frequency resolution and introduces significant noise into the spectrum. To save the power consumption of FFT, the multiplication with E3bT can be divided into two parts: first, numerical calculations are performed, followed by symbol calculations. The numerical calculation formula is shown in (2), and can be implemented using shift and addition operations, thereby reducing computational cost.

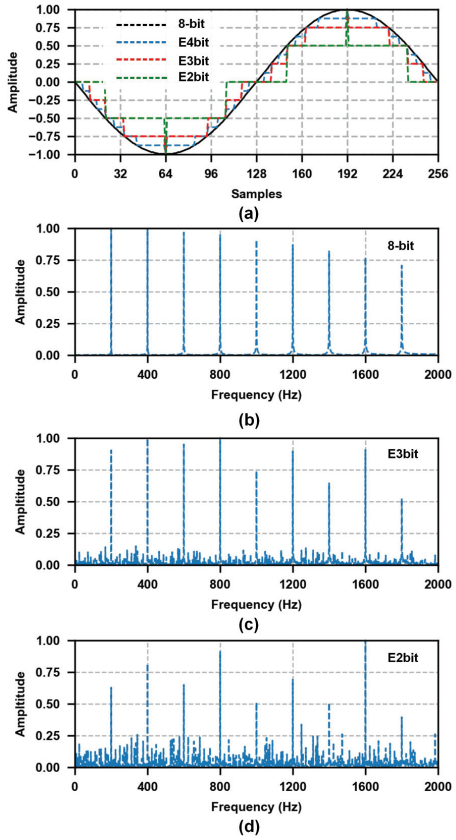


Fig. 4 (a) The quantized FFT twiddle, and the FFT results of multi-tone sine input range from 100 Hz to 2000 Hz using (b) 8-bit twiddle, (c) E3bT, and (d) E2bT.

$$T(x) = \begin{cases} 0, & |twiddle| = 0 \\ x >> 2, & |twiddle| = 0.25 \\ x >> 1, & |twiddle| = 0.5 \\ (x >> 2) + (x >> 1), & |twiddle| = 0.75 \\ x, & |twiddle| = 1 \end{cases} \tag{2}$$

The power operation calculates the modulus of the com-

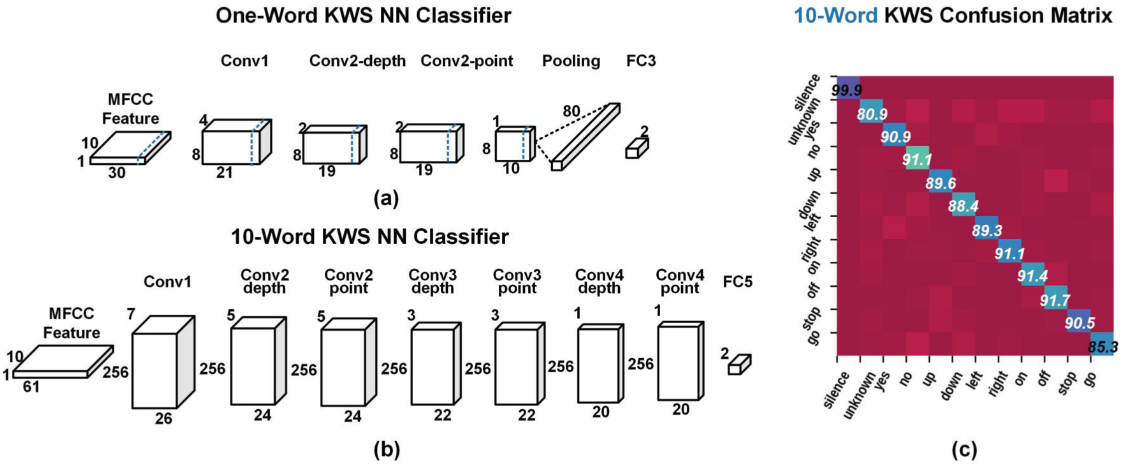


Fig. 5 (a) Neural network classifier architecture of 1-Word KWS and 10-Word KWS. (b) The simulated classification confusion matrix of 10-Word KWS.

Table I MFCC Computational Amounts Analysis

Operation	Theoretical MFCC			Multiplier-Free MFCC		
	Mult	**Add**	**Bit-Op**	**Mult**	**Add**	**Bit-Op**
Window	256	0	65536	0	0	0
FFT	2048	3072	622592	0	7168	114688
Power	256	128	69632	0	128	2048
Mel-Filter	208	208	59904	0	208	3328
DCT	400	400	115200	0	400	6500
Sum	3168	3808	932864	0	7904	126464

plex number spectrum from the FFT. We use Add-Power, as shown in (3), which replaces the modulus operations with the addition of absolute values, avoiding multiplication and square root operations on real numbers. Furthermore, a Mel-Filter with rectangular band-pass filter banks [12] is used instead of a triangular band-pass filter bank, and the E3bT is used instead of a cosine Twiddle in the DCT operation.

$$P(x) = |x_{real}| + |x_{image}| \tag{3}$$

Through optimization at each step, MF-MFCC replaces multiplication operations in the MFCC with addition operations and optimizes storage overhead. Table I shows that the number of bit operations for MF-MFCC is 13.5% (7.3x) of that for theoretical MFCC.

We trained 1-Word and 10-Word KWS classifiers with MF-MFCC using the Google Speech Commands Dataset (GSCD), as shown in Fig. 5(a). The 1-Word KWS, consisting of a 3-layer CNN and a 1-layer FC network with 1-bit weights, achieved 93.4% average accuracy on 'happy', 'dog', 'up', and 'down'. The 10-Word KWS, with 7 CNN layers and 1 FC layer, achieved 89.6% average accuracy on 'yes', 'no', 'up', 'down', 'left', 'right', 'on', 'off', 'stop', and 'go'. The confusion matrix is shown in Fig. 5(b).

To assess the impact of the MF-MFCC algorithm on the KWS system, we analyzed the 10-Word KWS accuracy using MF-MFCC compared to MFCC with single optimizations. As shown in Table II, window, Mel-filter, power, and DCT operations have minimal effects on accuracy. The E3bT-FFT slightly reduces accuracy but significantly lowers power consumption.

Table II 10-Word KWS accuracy with MF-MFCC

	Neural Network Accuracy (%)
Theoretical	94.04
Quantized Window	93.04
E3bit-Twiddle FFT	90.69
Add Power	92.51
Rectangle Mel-Filter	93.74
E3bit-Twiddle DCT	93.74
Multiplier-Free MFCC	89.73

3. Hardware Architecture

The KWS hardware architecture, shown in Fig. 6(a), includes an MFCC accelerator and a deep learning accelerator (DLA). The voice signal is stored in a ping-pong FIFO, and the MFCC accelerator computes the spectrum and transmits it to the deep learning accelerator. First, window and FFT operations are performed, followed by power, Mel-filter, and DCT operations. Power, Mel-Filter, and DCT share the same computing unit. The deep learning accelerator, consisting of 8 PEs and one batch normalization PE (BN-PE), determines whether keywords are detected based on the output of network.

The window function module includes a selector and shift logic. The window function value is determined based on the input audio index, followed by the corresponding shift operation.

The E3bT butterfly unit, shown in Fig. 6(b), consists of four E3bT multipliers and six adders. The E3bT multiplier splits multiplication into two parts: numerical and symbolic computation. First, the input is shifted or added based on the absolute value of the twiddle. Then, the numerical value is inverted if necessary, based on the product of the symbols, to produce the final output. As shown in Fig. 7(a), the frequency with an E3bT absolute value of 0.75 is 24.4%. Compared to 8-bit Twiddle multiplication, using E3bT re-

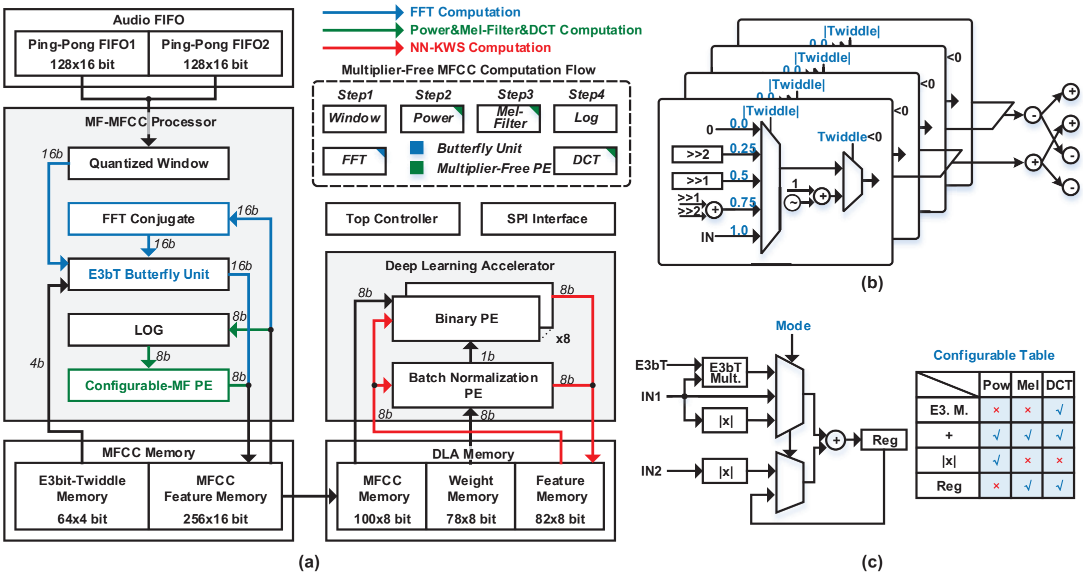


Fig. 6 (a) The hardware architecture of KWS, and the computation flow of MF-MFCC processor. (b) The E3bT butterfly unit. (c) The configurable multiplier-free processing element.

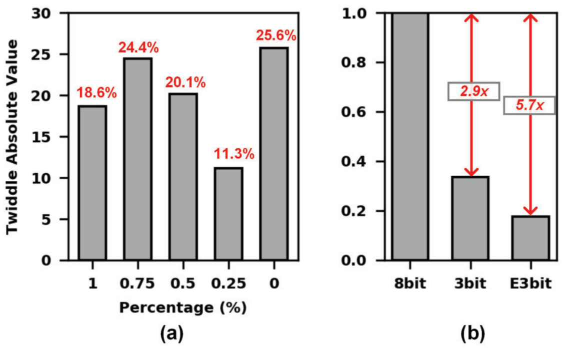
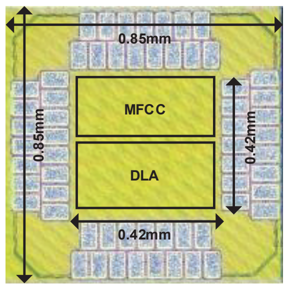


Fig. 7 (a) The E3bT usage statistics in 256-point FFT. (b) The normalization power of twiddle multiplier.



Process	65 nm
Area	0.85x0.85 mm^2
Memory	1.6 KB
Voltage	0.55 V
Frequency	125 KHz
Power	2.3 μW

Fig. 8 Die photograph and chip specifications.

duces multiplication power by 5.7x, as shown in Fig. 7(b).

The configurable multiplier-free processing element (MF-PE), shown in Fig. 6(c), consists of an adder, a MUX, and an E3bT multiplier. It performs calculations for Power, Mel, and DCT based on different configurations. For Power, the two inputs undergo absolute value operations and are then summed through an adder. For Mel, the input is accumulated using an adder and a register. In DCT operations, the input is first multiplied by E3bT, then accumulated with registers through an adder. Additionally, before DCT operations, the input undergoes logarithmic processing before being passed into the MF-PE.

The accelerator's calculation module includes a PE array and a BN-PE unit. The PE array contains 8 PEs. In binary neural networks, weights are quantized to 1 bit, representing 1 or -1, and activations are quantized to 8 bit. The binary PE replaces traditional multipliers with MUXs and inverters, feeding the results into an adder for accumulation. After calculations in each CNN and FC layer, BN is applied using the BN-PE unit. The weights of BN operation are 8-bit fixed-point data and it require an 8-bit multiplier for execution.

4. Measurement Results

The KWS chip is implemented using a 65-nm CMOS process with a core area of 0.17 mm^2, as shown in Fig. 8. The on-chip memory is 1,681 bytes, with 1,188 bytes allocated to the MF-MFCC accelerator and 493 bytes to the deep learning accelerator. To reduce power consumption, on-chip memory is implemented using latches. The test platform uses an FPGA to generate the 8 KHz audio signal to digital KWS chip. The initial memory value is transferred from the PC to the PCB via SPI.

The accuracy of the KWS chips was tested using the GSCD dataset. For the one-word KWS chip, four keywords—'happy', 'dog', 'up', and 'down' were tested separately, with other keywords in GSCD used as filters. During testing, all keywords and filters were randomly concatenated, and silence was added at a 1:6 ratio to create a 1-hour audio sample. Fig. 9(a) shows the accuracy of one-word detec-

Table III Performance Comparison Table

Audio Feature Extractor	VLSI'19 [16]	ISSCC'20 [15]	JSSC'21 [34]	ISSCC'22 [26]	JSSC'23 [14]	JSSC'23 [33]	JSSC'24 [35]	This Work	
Technology (nm)	65	28	180	65	28	65	28	65	
FEx Technique	MFCC	MFCC	IROC	ASP	MFCC	ASP	MFCC	MF-MFCC	
FFT Method	Radix-2 FFT	Piplined FFT	N/A	N/A	Radix-2 Real FFT	N/A	Piplined FFT	Radix-2 Real E3bT-FFT	
Frequency Range (Hz)	N/A	20-4k	N/A	110-10.4k	20-4k	100-4k	8k	20-4k	
Frame Shift (ms)	16	16	N/A	16	16	10	16	16	
Audio Bit-width (bits)	10	8	N/A	N/A	8	8	10	16	
Analog Front-End	Yes	No	Yes	Yes	Yes	Yes	Yes	No	
FFT Computation Power (nW)	N/A	205	N/A	N/A	130	N/A	56.9	64	
FEx Computation Power (nW)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	76	
FEx Memory Power (nW)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1695	
FEx Total Power (nW)	7137	336	35	11040	180	80	111.6	1771	
Keyword Spotting Demonstrations									
Accuracy on GSCD	90.8% on 10 words	97.3% on 1 word	94.0% on 1 words	86% on 10 words	92.8% on 5 words	91.5% on 10 words	98.1% on 1 word	94.9% on 1 word	89.6% on 10 words
Classifier	on-chip RNN	on-chip DS-CNN	on-chip CNN	on-chip RNN	on-chip Skip RNN	off-chip CNN	on-chip TD-SNN	on-chip DS-CNN	off-chip DS-CNN
Voltage (V)	0.6-1.2	0.41	0.6	0.5	N/A	N/A	0.36	0.55	N/A
DLA Power (μW)	8.97	0.17	0.064	11.3	1.3	N/A	0.50	0.53 (1-Word)	N/A
Total Power (μW)	16.1	0.51	0.148	32.3	1.48	N/A	0.61	2.3 (1-Word)	N/A
Chip Area (mm^2)	2.56	0.23	N/A	2.03	0.8	N/A	0.47	0.17	N/A

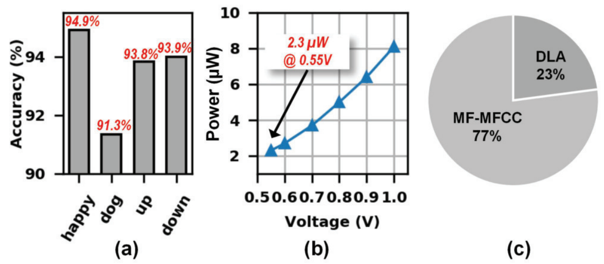


Fig. 9 (a) Measured 1-Word KWS accuracy on GSCD dataset. (b) Chip power versus voltage of one typical chip. (c) Chip power consumption breakdown.

tion. It achieved 94.9% accuracy on the GSCD dataset. At a 0.55 V supply voltage, the chip's power consumption is 2.3 μW, as shown in Fig. 9(b). The power consumption and breakdown of each module are detailed in Fig. 9(c). The MFCC computing power accounts for only 3.3% of the chip's total power consumption, while the overall computing power of the MFCC accounts for 77%. Compared to previous academic work, as shown in Table III.

5. Conclusion

An ultra-low-power KWS chip is fabricated in 65 nm CMOS with a 0.17 mm^2 area. An E3bT FFT and MF-MFCC are used to reduce computations by 7.3x. A KWS processor with a configurable MF-PE further reduces power and area. The chip consumes 2.3 μW at a 0.55 V supply voltage, with the MFCC computation power at 76 nW, accounting for only 3.3% of the total chip power. The power of MFCC is reduced by 4x than [16].

Acknowledgments

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